EcoRoute — Database Architecture & Schema Specification

Complete PostgreSQL Prisma Schema (16 Models, Enums, Relationships & Data Flow Explanation)

1. Executive Database Overview

The **EcoRoute** database architecture is built on top of **PostgreSQL** using **Prisma ORM** as the primary data abstraction layer. It defines **16 relational models**, **6 enums**, and strict foreign key constraints guaranteeing high data integrity for e-waste tracking, user identity management, doorstep collection dispatch, and CPCB regulatory compliance.

- **Primary Database:** PostgreSQL managed via Prisma ORM (`prisma/schema.prisma`).
- **Cloud Realtime Backup:** Supabase Client integration (`lib/supabase.ts`) for realtime notifications and cloud asset storage.
- **Offline Disk Fallback:** Local JSON storage (`data/users.json` & `lib/user-store.ts`) allowing demo execution without an active database URL.
- **Primary Keys:** Universally Unique Identifiers (UUID v4) across all tables (`@default(uuid())`).
- **Cascading Deletes:** Automatic relational cleanup on profile, e-waste items, rewards, and notification deletions (`onDelete: Cascade`).

2. Database Enums Reference

Enum Name	Allowed Values	Business Purpose & Scope
`Role`	CITIZEN, OFFICER, ADMIN	Role-based access control (Citizens, Field Officers, Central Admins).
`PickupStatus`	SUBMITTED, APPROVED, ASSIGNED, COLLECTED, DELIVERED, RECYCLED, COMPLETED, CANCELLED	8-stage lifecycle tracking for doorstep e-waste collection requests.
`Priority`	NORMAL, HIGH, URGENT, CRITICAL	Urgency tagging for hazardous items and dumping complaints.
`HazardLevel`	LOW, MODERATE, HIGH, CRITICAL	Environmental hazard rating based on battery/mercury content.
`NotificationType`	PICKUP, NOTICE, REWARD, REMINDER, SYSTEM	Classification for system alerts and activity notifications.
`ComplaintStatus`	OPEN, IN_PROGRESS, RESOLVED, ESCALATED	Grievance tracking for illegal e-waste dumping reports.

3. Detailed Table & Model Explanation (All 16 Models)

1. User (`users`) — *Stores core authentication accounts.*

Field Name	Data Type & Constraint	Description & Relational Flow
`id`	String (UUID, PK)	Unique user identification identifier.
`email`	String (Unique)	Account login email address.
`passwordHash`	String	Bcrypt multi-round hashed password string.
`role`	Role (Enum)	CITIZEN (Default), OFFICER, or ADMIN access level.
`Relations`	Profile, Pickups	Has 1 Profile, N Pickups, N Rewards, N Certificates, N Complaints, 1 Driver/Officer/Admin.

2. Profile (`profiles`) — *Citizen user profile and identity credentials.*

Field Name	Data Type & Constraint	Description & Relational Flow
`userId`	String (FK, Unique)	Foreign key referencing `users.id` (On Delete Cascade).
`fullName / phone`	String	Citizen contact full name and phone number.
`citizenId`	String (Unique)	Government Citizen Identification Code (e.g. `DL-2026-8941`).
`aadhaarLinked`	Boolean	Flags verified Aadhaar authentication status.

3. PickupRequest (`pickup_requests`) — *Core doorstep collection request.*

Field Name	Data Type & Constraint	Description & Relational Flow
`requestId`	String (Unique)	Unique tracking code (e.g. `REQ-2026-8941`).
`status`	PickupStatus (Enum)	Active status in pickup lifecycle (`SUBMITTED` to `COMPLETED`).
`hazardLevel`	HazardLevel (Enum)	AI-tagged hazard rating (LOW, MODERATE, HIGH, CRITICAL).
`centerId / driverId`	String (FK)	Assigned Recycling Center, Driver, and Vehicle IDs.

4. EWasteItem (`ewaste_items`) — *Individual electronic scrap item payload.*

Field Name	Data Type & Constraint	Description & Relational Flow
`pickupRequestId`	String (FK)	Parent pickup request ID (On Delete Cascade).
`deviceName / brand`	String	Device model name, brand, and category.
`imageUrl`	String?	Cloudinary photo URL or Base64 scan image string.

5. RecyclingCenter (`recycling_centers`) — *CPCB-certified recycling facility registry.*

Field Name	Data Type & Constraint	Description & Relational Flow
`registrationNo`	String (Unique)	Official CPCB license registration number.
`capacityTonnes`	Float	Monthly max recycling processing load capacity.
`latitude / longitude`	Float	GPS coordinates for Leaflet GIS map rendering.

6. Vehicle (`vehicles`) — *EV collection fleet van registry.*

Field Name	Data Type & Constraint	Description & Relational Flow
`vehicleNumber`	String (Unique)	EV registration plate number.
`assignedDriverId`	String (FK, Unique)	Assigned field collection driver ID.
`batteryFuelPercent`	Int	Real-time battery percentage telemetry.

7. Driver (`drivers`) — *Field collection driver profile.*

Field Name	Data Type & Constraint	Description & Relational Flow
`userId`	String (FK, Unique)	Foreign key referencing `users.id`.
`district / rating`	String / Float	Operating district zone and performance rating.

8. Officer (`officers`) — *CPCB regional environmental field officer.*

Field Name	Data Type & Constraint	Description & Relational Flow
`officerCode`	String (Unique)	Unique officer badge code (e.g. `OFF-DEL-104`).
`dutyZone`	String	Assigned regional environmental inspection zone.

9. Admin (`admins`) — *Central CPCB government executive account.*

Field Name	Data Type & Constraint	Description & Relational Flow
`adminCode`	String (Unique)	Executive nodal director access code.
`roleTitle`	String	Department executive designation.

10. Certificate (`certificates`) — *Digital Green Recycling & Carbon Offset Proof.*

Field Name	Data Type & Constraint	Description & Relational Flow
`certificateNo`	String (Unique)	Official certificate code (e.g. `CERT-DL-8902`).
`co2SavedKg`	Float	Calculated CO2 offset saved (Weight kg x 1.45).
`cpbLicenseNo`	String	Issuing facility CPCB licensing badge.

11. Complaint (`complaints`) — *Citizen illegal e-waste dumping report.*

Field Name	Data Type & Constraint	Description & Relational Flow
`complaintId`	String (Unique)	Grievance ticket ID.
`assignedOfficerId`	String (FK)	Assigned CPCB field inspection officer ID.

12. AIClassification (`ai_classifications`) — *AI computer vision classification log.*

Field Name	Data Type & Constraint	Description & Relational Flow
`confidenceScore`	Float	AI identification confidence percentage (e.g. 98.4%).
`estimatedValue`	String	Extracted market scrap valuation range.

13-16. Auxiliary Models — *Notifications, Rewards, Announcements, Logs.*

Field Name	Data Type & Constraint	Description & Relational Flow
`Reward`	Points Catalog	Citizen green points balance and vouchers.
`Notification`	Alert System	Real-time system notification stream.
`ActivityLog`	Audit Trail	Security tracking log storing action details and IP address.

4. Complete Raw Prisma Schema Code (`schema.prisma`)

Below is the exact, complete source code of the production Prisma schema located at `prisma/schema.prisma`:

```
// ============================================================
// EcoRoute — Prisma Schema (PostgreSQL)
// Government AI Powered E-Waste Management Platform
// ============================================================

generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

// ■■ Enums ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
enum Role {
  CITIZEN
  OFFICER
  ADMIN
}

enum PickupStatus {
  SUBMITTED
  APPROVED
  ASSIGNED
  COLLECTED
  DELIVERED
  RECYCLED
  COMPLETED
  CANCELLED
}

enum Priority {
  NORMAL
  HIGH
  URGENT
  CRITICAL
}

enum HazardLevel {
  LOW
  MODERATE
  HIGH
  CRITICAL
}

enum NotificationType {
  PICKUP
  NOTICE
  REWARD
  REMINDER
  SYSTEM
}

enum ComplaintStatus {
  OPEN
  IN_PROGRESS
  RESOLVED
  ESCALATED
}

// ■■ 1. User Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model User {
  id               String   @id @default(uuid())
  email            String   @unique
  passwordHash     String
  role             Role     @default(CITIZEN)
  verified         Boolean  @default(false)
  verificationToken String?
```

```
  resetToken       String?

  resetTokenExpiry DateTime?
  createdAt        DateTime @default(now())
  updatedAt        DateTime @updatedAt

  profile Profile?

  // Relations
  pickups       PickupRequest[]
  notifications Notification[]
  rewards       Reward[]
  certificates  Certificate[]
  complaints    Complaint[]
  activityLogs  ActivityLog[]
  driver        Driver?
  officer       Officer?
  admin         Admin?

  @@map("users")
}

// ■■ 2. Profile Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Profile {
  id            String   @id @default(uuid())
  userId        String   @unique
  user          User     @relation(fields: [userId], references: [id], onDelete: Cascade)
  fullName      String
  phone         String
  address       String
  city          String
  state         String
  pinCode       String
  citizenId     String   @unique
  aadhaarLinked Boolean  @default(false)
  avatarUrl     String?
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt

  @@map("profiles")
}

// ■■ 3. Pickup Request Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model PickupRequest {
  id              String       @id @default(uuid())
  requestId       String       @unique
  userId          String
  user            User         @relation(fields: [userId], references: [id])
  deviceName      String
  category        String
  brand           String?
  requestDate     DateTime     @default(now())
  pickupDate      String
  timeSlot        String
  status          PickupStatus @default(SUBMITTED)
  priority        Priority     @default(NORMAL)
  address         String
  city            String
  district        String
  pinCode         String
  estimatedWeight Float?
  instructions    String?
  contactMethod   String?      @default("SMS & Call")
  rewardPoints    Int          @default(0)
  hazardLevel     HazardLevel  @default(LOW)

  centerId        String?
  center          RecyclingCenter? @relation(fields: [centerId], references: [id])
  driverId        String?
  driver          Driver?          @relation(fields: [driverId], references: [id])
  vehicleId       String?
  vehicle         Vehicle?         @relation(fields: [vehicleId], references: [id])

  ewasteItems     EWasteItem[]
  certificate     Certificate?
  createdAt       DateTime         @default(now())
```

```
  updatedAt       DateTime         @updatedAt

  @@map("pickup_requests")
}

// ■■ 4. E-Waste Item Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model EWasteItem {
  id              String        @id @default(uuid())
  pickupRequestId String
  pickupRequest   PickupRequest @relation(fields: [pickupRequestId], references: [id], onDelete: Cascade)
  deviceName      String
  brand           String?
  category        String
  condition       String
  estimatedAge    String?
  description     String?
  imageUrl        String?
  createdAt       DateTime      @default(now())

  @@map("ewaste_items")
}

// ■■ 5. Recycling Center Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model RecyclingCenter {
  id                String   @id @default(uuid())
  name              String
  registrationNo    String   @unique
  address           String
  city              String
  state             String
  pinCode           String
  phone             String
  email             String
  workingHours      String
  rating            Float    @default(4.5)
  reviewCount       Int      @default(0)
  distanceKm        Float    @default(0)
  capacityTonnes    Float    @default(500)
  monthlyLoadTonnes Float    @default(0)
  todayLoadKg       Float    @default(0)
  isOpenNow         Boolean  @default(true)
  status            String   @default("Operational")
  latitude          Float
  longitude         Float
  createdAt         DateTime @default(now())
  updatedAt         DateTime @updatedAt

  pickups           PickupRequest[]
  aiClassifications AIClassification[]

  @@map("recycling_centers")
}

// ■■ 6. Vehicle Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Vehicle {
  id                String   @id @default(uuid())
  vehicleNumber     String   @unique
  type              String   @default("EV Van")
  assignedDriverId  String?  @unique
  district          String
  currentRoute      String
  status            String   @default("Idle")
  batteryFuelPercent Int     @default(100)
  lastMaintenance   String
  createdAt         DateTime @default(now())
  updatedAt         DateTime @updatedAt

  driver            Driver?  @relation(fields: [assignedDriverId], references: [id])
  pickups           PickupRequest[]

  @@map("vehicles")
}

// ■■ 7. Driver Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Driver {
```

```
  id                  String   @id @default(uuid())
  userId              String   @unique
  user                User     @relation(fields: [userId], references: [id], onDelete: Cascade)
  driverName          String
  phone               String
  district            String
  availability        String   @default("Available")
  rating              Float    @default(4.8)
  assignedPickupsCount Int     @default(0)
  completedToday      Int      @default(0)
  createdAt           DateTime @default(now())
  updatedAt           DateTime @updatedAt

  vehicle             Vehicle?
  pickups             PickupRequest[]

  @@map("drivers")
}

// ■■ 8. Officer Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Officer {
  id          String   @id @default(uuid())
  userId      String   @unique
  user        User     @relation(fields: [userId], references: [id], onDelete: Cascade)
  officerCode String   @unique
  department  String   @default("CPCB Regional Operations")
  dutyZone    String
  status      String   @default("Shift Active")
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt

  complaints  Complaint[]

  @@map("officers")
}

// ■■ 9. Admin Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Admin {
  id         String   @id @default(uuid())
  userId     String   @unique
  user       User     @relation(fields: [userId], references: [id], onDelete: Cascade)
  adminCode  String   @unique
  roleTitle  String   @default("CPCB Nodal Director")
  department String   @default("MoEFCC Central Executive")
  createdAt  DateTime @default(now())

  @@map("admins")
}

// ■■ 10. Notification Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Notification {
  id          String           @id @default(uuid())
  userId      String
  user        User             @relation(fields: [userId], references: [id], onDelete: Cascade)
  title       String
  message     String
  type        NotificationType @default(SYSTEM)
  read        Boolean          @default(false)
  actionUrl   String?
  actionLabel String?
  createdAt   DateTime         @default(now())

  @@map("notifications")
}

// ■■ 11. Reward Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Reward {
  id              String    @id @default(uuid())
  userId          String
  user            User      @relation(fields: [userId], references: [id], onDelete: Cascade)
  title           String
  description     String
  iconName        String
  pointsRequired  Int       @default(100)
  progressPercent Int       @default(0)
```

```
  isUnlocked      Boolean   @default(false)
  unlockedAt      DateTime?
  createdAt       DateTime  @default(now())

  @@map("rewards")
}

// ■■ 12. Certificate Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Certificate {
  id              String        @id @default(uuid())
  userId          String
  user            User          @relation(fields: [userId], references: [id])
  pickupRequestId String        @unique
  pickupRequest   PickupRequest @relation(fields: [pickupRequestId], references: [id])
  certificateNo   String        @unique
  issueDate       String
  recycledDevice  String
  category        String
  weightKg        Float
  co2SavedKg      Float
  recyclerName    String
  cpbLicenseNo    String
  downloadUrl     String?
  createdAt       DateTime      @default(now())

  @@map("certificates")
}

// ■■ 13. Complaint Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Complaint {
  id               String          @id @default(uuid())
  complaintId      String          @unique
  userId           String
  user             User            @relation(fields: [userId], references: [id])
  category         String
  priority         Priority        @default(NORMAL)
  assignedOfficerId String?
  assignedOfficer  Officer?        @relation(fields: [assignedOfficerId], references: [id])
  status           ComplaintStatus @default(OPEN)
  description      String
  dateFiled        DateTime        @default(now())

  @@map("complaints")
}

// ■■ 14. Announcement Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model Announcement {
  id          String   @id @default(uuid())
  title       String
  category    String
  target      String   @default("All Portals")
  description String
  dateText    String
  publishedAt DateTime @default(now())

  @@map("announcements")

}

// ■■ 15. AI Classification Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model AIClassification {
  id                   String          @id @default(uuid())
  imageUrl             String
  deviceCategory       String
  confidenceScore      Float
  estimatedValue       String
  hazardLevel          HazardLevel     @default(MODERATE)
  hazardDescription    String
  disposalRecommendation String
  suggestedCenterId    String?
  suggestedCenter      RecyclingCenter? @relation(fields: [suggestedCenterId], references: [id])
  createdAt            DateTime        @default(now())

  @@map("ai_classifications")
}
```

```
// ■■ 16. Activity Log Model ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
model ActivityLog {
  id        String   @id @default(uuid())
  userId    String?
  user      User?    @relation(fields: [userId], references: [id], onDelete: SetNull)
  action    String
  details   String
  ipAddress String?
  createdAt DateTime @default(now())

  @@map("activity_logs")
}
```

End of Official Database Documentation — EcoRoute Platform Specification